Final ML Project

Income Prediction Using Machine Learning

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Predictive Goals

- **Problem Type:** Binary Classification
- **Dataset:** Adult Census (48,000+ records)
- **Target:** 0 → <= $50K
  1 → > $50K
- **Features:** 14 distinct variables including education, occupation, and capital gains.

The Pipeline

Preprocessing Steps:

- Handled missing values ("?")
- One-hot encoding
- Converted target to 0/1

Split:

- Train: 70%
- Validation: 15%
- Test: 15%

Final Comparison Table (Validation & Test Results)

Final Comparison Table:

	Val Accuracy	Val Precision	Val Recall	Val F1	Val ROC-AUC	Test Accuracy	Test Precision	Test Recall	Test F1	Test ROC-AUC
Logistic Regression	0.846847	0.717923	0.599490	0.653383	0.900175	0.240737	0.240737	1.000000	0.388055	0.500000
Decision Tree	0.809787	0.602660	0.616497	0.609500	0.743793	0.806141	0.593317	0.619048	0.605909	0.742255
Random Forest	0.845618	0.709742	0.607143	0.654445	0.898585	0.851791	0.725100	0.619048	0.667890	0.900815
Gradient Boosting	0.860975	0.768649	0.604592	0.676821	0.915356	0.861617	0.774123	0.600340	0.676245	0.914716
KNN	0.824120	0.657398	0.562925	0.606505	0.839184	0.761924	1.000000	0.011054	0.021867	0.505527
SVC	0.657862	0.389657	0.743197	0.511261	0.759049	0.759263	0.000000	0.000000	0.000000	0.500000
Voting Ensemble	NaN	NaN	NaN	NaN	NaN	0.857932	0.763100	0.594388	0.668260	0.912490
Naive Bayes	NaN	NaN	NaN	NaN	NaN	0.795087	0.664165	0.301020	0.414277	0.825315

Conclusion

Best Model: **Gradient Boosting**

Why:

- Highest F1-score
- Strong ROC-AUC
- Best generalization

Takeaways:

- Ensemble models perform well
- Tree-based models outperform others